

Skills Summary

Similar Triangles: Sides, Perimeter and Area

Use this reference while completing your worksheet or when studying.

The one rule everything below comes from. If two figures are similar with scale factor k (target length over starting length), then every *length* is multiplied by k , every *perimeter* is multiplied by k , and every *area* is multiplied by k^2 .

Skill 1 — Finding a missing side

Example. $\triangle ABC \sim \triangle DEF$ with $AB = 4$ cm, $BC = 9$ cm and $DE = 6$ cm. Find EF .

Step 1. Want EF ; I have the pair AB with DE to build the ratio and the side BC that matches EF — one proportion, first triangle on top on both sides.

$$\text{Step 2. } \frac{4}{6} = \frac{9}{EF} \Rightarrow EF = 9 \cdot \frac{6}{4}$$

$$EF = 13.5 \text{ cm}$$

Try it. $\triangle ABC \sim \triangle DEF$ with $AB = 10$ cm, $BC = 6$ cm and $DE = 25$ cm. Find EF .

check: $EF = 15$ cm

Skill 2 — Finding the scale factor

Example. Two similar triangles have matching sides of 4 cm and 18 cm. Find the scale factor from the smaller to the larger.

Step 1. Want k going small to large, and I have one matching pair — so divide the target length by the starting length.

$$\text{Step 2. } k = \frac{18}{4}$$

$$k = 4.5$$

Try it. Two similar triangles have matching sides of 7 cm and 28 cm. Find the scale factor from the smaller to the larger.

check: $k = 4$

Skill 3 — Perimeter scales like a side

Example. $\triangle ABC \sim \triangle DEF$ with $AB = 5$ cm and $DE = 8$ cm. The perimeter of $\triangle ABC$ is 20 cm. Find the perimeter of $\triangle DEF$.

Step 1. Want a perimeter, and a perimeter is just three lengths added — so it scales by k itself, never by k^2 .

$$\text{Step 2. } k = \frac{8}{5}, \text{ so } P = 20 \cdot \frac{8}{5}$$

$$P = 32 \text{ cm}$$

Try it. $\triangle ABC \sim \triangle DEF$ with $AB = 6$ cm, $DE = 9$ cm, and the perimeter of $\triangle ABC$ equal to 22 cm. Find the perimeter of $\triangle DEF$.

check: $P = 33$ cm

Skill 4 — Area scales like the square

Example. Two similar triangles have matching sides 3 cm and 7 cm. The smaller has area 18 cm^2 . Find the area of the larger.

Step 1. Want an area; a triangle stretches in two directions at once, so the factor is applied twice — multiply by k^2 .

Step 2. $k = \frac{7}{3}$, so $A = 18 \left(\frac{7}{3}\right)^2$
 $A = 98 \text{ cm}^2$

Try it. Two similar triangles have matching sides 5 cm and 8 cm. The smaller has area 25 cm^2 . Find the area of the larger.

check: $A = 98 \text{ cm}^2$

(!) Watch out: going *backwards* from areas, take the square root first. If the areas are in the ratio $9 : 4$, the sides and perimeters are in the ratio $3 : 2$, not $9 : 4$.